

Midterm Study Guide

Prof. Sirsha Pattanayak · Session 1 · Waiting & Queuing · Revenue & Yield Management · Service Facility Layout (slide 47 onwards)

How to use this in the last days before the exam

1. Read **Part 0** (formula sheet map + traps) once. It is the highest-value page.
2. For each part: skim the formula boxes, then **cover the solution and redo every solved example** on paper. Numbers here were all re-computed and checked.
3. In the exam, always follow the same loop: **identify the model** → **list givens with units** → **pick formulas** → **compute** → **interpret / decide**.

Contents

1. Part 0 — Formula sheet: what is on it, what is not, and the traps
2. Part 1 — Session 1: Nature of services (theory)
3. Part 2 — Waiting lines & queuing (formulas + 10 solved examples)
4. Part 3 — Revenue & yield management (formulas + 7 solved examples)
5. Part 4 — Service facility layout (formulas + 4 solved examples)
6. Part 5 — 2-hour exam plan, unit checks, last-minute checklist

Topic	Emphasis (my estimate from your note)	Type
Waiting lines (M/M/1, M/M/1/N, M/M/c, M/D/1, cost trade-offs)	Largest	Numerical
Revenue mgmt (Littlewood, overbooking, multi-segment pricing)	Large	Numerical + short concept
Facility layout (line balancing, utilization, bottleneck, cost)	Medium	Numerical
Session 1 + game theory in RM + servicescape/layout concepts	Small	Short answers

PART 0

The Formula Sheet: how to use it

The sheet has 3 pages. Know exactly what is on it and what is missing.

Formula-sheet block	Use for	Watch out
M/M/1/∞/∞ (11 items)	Single server, unlimited queue and population. Needs $\lambda < \mu$.	Items 6&7 are Little's law. Item 5: $L_S = L_Q + \rho$. Item 11 has a typo ("==") — it is $W_Q = \lambda/[\mu(\mu-\lambda)]$.
M/M/1/N/∞	Single server, max N customers in the <i>system</i> (waiting room + server). Blocked customers are lost.	Use λ_{eff} (not λ) in W_S , W_Q . Formula for L_S fails when $\rho=1$ (not expected).
M/M/c/∞/∞	c identical servers, one queue.	$\rho = \lambda/(c\mu)$ in P_0 , but in the L_Q formula the symbol ρ means $r = \lambda\mu$ (offered load). See trap below.
M/D/1/∞/∞	Constant (fixed) service time: automatic machines, car wash, compactor.	L_Q and W_Q are exactly half of the M/M/1 values (extra 2 in denominator).
Page 3 (bottom): capacity & cost lines	Layout (line balancing) and the critical ratio for Littlewood/overbooking.	$C_U = p-c$ and $C_0 = c-v$ are the <i>generic newsvendor</i> definitions. For RM, redefine them (see Part 3).

TRAP 1 — P IN M/M/C. On the sheet, $\rho = \lambda/(c\mu)$ is *utilisation*. Inside $L_Q = \rho^{c+1}P_0 / [(c-\rho)^2(c-1)!]$ the ρ is really $\lambda\mu$. Check with $c=1$: it must collapse to $\lambda^2/[\mu(\mu-\lambda)]$. Only that reading passes. Also in P_0 , $(c\rho)^n = (\lambda\mu)^n$, so it is consistent.

TRAP 2 — UNITS. Convert everything to **per hour** first. "Customers arrive every 3 minutes" $\Rightarrow \lambda = 60/3 = 20$ per hour. Formulas return hours; multiply by 60 for minutes.

TRAP 3 — NOT ON THE SHEET (MEMORISE!). $P(n > k \text{ in system}) = \rho^{k+1} (M/M/1) \cdot$ Probability an arrival waits in M/M/c • Littlewood: $P(D_H > Q) = F_L/F_H \cdot$ optimal price $p = A/(2B) + c/2 \cdot$ capacitated pricing with Lagrange multiplier • utilisation of bottlenecked lines with cost per customer • normal **z-values** (table below).

WHICH LINE OF THE SHEET ANSWERS WHICH QUESTION

"% of time server idle" $\rightarrow P_0$ "% of time no queue" $\rightarrow P_0 + P_1$ (server has at most 1 customer)

"Average number waiting" $\rightarrow L_Q$ "in the system" $\rightarrow L_S$

"Average time waiting to be served" $\rightarrow W_Q$ "total time incl. service" $\rightarrow W_S$

"Server utilisation / busy" $\rightarrow \rho$ (or $1-P_0$) "lost customers" $\rightarrow \lambda \cdot P_N$

"How many seats to protect / how many to overbook" $\rightarrow CR$ and $Q^* = \mu + z\sigma$

"Capacity / cycle time / utilisation of a line" $\rightarrow$ page-3 capacity lines

NORMAL Z-TABLE SHORTCUTS (FOR CR $\rightarrow$ Z)

CR = $P(X \leq Q)$	0.10	0.20	0.2857	0.29	0.30	0.40	0.50	0.60	0.70	0.75	0.80	0.90	0.95	0.975	0.99
z	-1.28	-0.84	-0.57	-0.55	-0.52	-0.25	0	0.25	0.52	0.67	0.84	1.28	1.64	1.96	2.33

Symmetry: $z(1-p) = -z(p)$. If the exam gives its own z-table, use that; small differences in the 2nd decimal are fine.

Session 1: Nature of Services

Mostly theory — expect short answers / 1-2 marks. Learn the lists and the examples.

CORE IDEAS

- **Definition by exclusion:** services are defined by what they are *not* — economic reports call anything not “goods-producing” or “extraction-based” a service (retail, wholesale, transport, finance, lodging, education, government, entertainment...).
- **Evolution:** Pre-industrial (domestic servants, sailors; relationships; little tech) → Industrial (assembly line, human as a “cog”, output focus, blue- vs white-collar) → Post-industrial (health, education, recreation; information rather than muscle; far higher share of service activity).
- **Service–product continuum:** product-dominant (passenger cars, machine tools) ↔ mixed (restaurants, fitness centres, hospitals) ↔ service-dominant (consulting, legal, logistics, tourism, facilities maintenance).

SERVICES VS MANUFACTURING (THE 5 DIFFERENCES)

Feature	Manufacturing	Services
Customer role	Not part of the process	Customer is an input; participates (facility design, decor, layout, noise matter; open kitchens build confidence).
System type	Closed system	Permeable system
Simultaneity	Produce, store, then sell	Created and consumed at the same time
Inventory	Inventory control	Cannot be inventoried → queuing / capacity management instead
Perishability	Stockable	Perishable (empty seat / room / hour is lost forever) → revenue management
Tangibility	Customer can see, touch, test	Intangible; performance judged after/while consumed

CLASSIFICATIONS OF SERVICES

Type	Distinctive	Example	Main managerial issue
B2B	Customer is an organisation; buyer may not be end user	Consultancy, maintenance	Demonstrate value, reliability, customisation
B2C	Individual is buyer/user	Hotel, bank, retail	Consistency across heterogeneous customers
Internal	Customer is another unit in same firm	HR, IT, Finance	Show value vs outsourcing
Public (G2C/G2B)	Little choice; tax-funded	Govt hospital, school, prisons	Access, fairness, acceptable quality, capacity
Not-for-profit	Beneficiary may not be payer	Faith orgs, aid agencies	Mission effectiveness

Customer-contact classification

- **High contact = pure services** (hospitals, restaurants, banking) → **mixed** → **Low contact = quasi-manufacturing** (distribution centres, warehouses).

Service process matrix (labour intensity vs interaction & customisation)

	Low interaction & customisation	High interaction & customisation
Low labour intensity	Service factory — airlines	Service shop — hospital
High labour intensity	Mass service — retailing	Professional service — consultancy

Challenges by cell: low-labour/low-customisation (service factory) → capital decisions, technology, demand peaks, scheduling. High-labour/high-customisation (professional) → hiring & training, SOPs, workforce scheduling, control of far-flung locations, making service “warm”, quality vs cost, customers intervening in the process.

OTHER SLIDE POINTS

- **Low-cost vs full-service airline:** high volume/low cost, short hauls, secondary airports, no frills vs global network, long haul with partner connections, primary airports, economy-to-first cabins.
- **Service blueprint:** strategic elements (service concept, service design) vs tactical/operational elements (facility, revenue management, queuing theory).
- **Sharing economy:** classical inventory asks *how much / when to order* (controlled supply). On platforms supply and demand stimulate each other and are not controlled by the platform → the decision becomes **matching** a unit of supply to a unit of demand.
- **Internet service design elements:** product, process, touch point, outcome; product/technology/task; customer/employee; performance.
- **New services (examples):** AlaaS, carbon-footprint auditing, elder companionship, pet tech, digital detox (fast growth); circular economy, urban farming, skill barter, therapy bots (scalable); metaverse, space tourism, memory preservation (transformational).

Waiting Lines & Queuing

The biggest numerical block. Master M/M/1, M/M/1/N, M/M/c, M/D/1 and the cost trade-off.

2.1 CONCEPTS (1-2 MARK THEORY)

- **Why lines form:** (a) workload > capacity, or (b) **variability** in arrivals and/or service times — even when average workload < capacity. Class demo: single teller, workload 50 min vs 60 min available → total wait **52 min**; two tellers (120 min available) → **7 min**. So **waiting is not linear in capacity**.
- **Balking** = arrival sees long queue and does not join. **Reneging** = leaves the queue before service. **Jockeying** = switches queues. (Single queue → no jockeying; multiple queues → jockeying.)
- **Channel** = number of servers; **phase** = number of sequential steps. Single-channel/single-phase (one cashier); single-channel/multi-phase (car wash: wash → dry); multi-channel/single-phase (bank counters); multi-channel/multi-phase (hospital: registration → doctor → tests).
- **Kendall notation M/M/c/N/pop:** arrivals Poisson (M), service Exponential (M) or Deterministic (D), c servers, N = max in system, population size.
- **Economics:** cost of capacity rises with servers, cost of waiting falls; **total cost is U-shaped**; minimum = optimal capacity. Waiting costs = opportunity cost of time, psychological cost, abandonment.
- **Stability:** for infinite queue models need $\rho < 1$.

Psychology of waiting (tactics)

Law of service	Tactic	Example
Unoccupied time feels longer	Distract / entertain	Mirrors in Boston hotel elevator lobby instead of more elevators
Pre-process waits feel longer than in-process	Get customers "in process" quickly	Menus, welcome drinks; fill a form while waiting
Uncertain waits feel longer than known waits	Communicate expected wait upfront and update	Call-centre "expected wait 6 minutes"
Fairness	Tokens, FCFS, one common queue, transparent priority	Car dealership: appointment customers have priority, walk-ins get token
Meaningful wait	Let customers do something useful	Fill forms, choose from menu
Social comparison	Show queue position and progress	"You are #3", A21 → A22 → A23

Managing demand

- Challenges: cyclical demand (daily/weekly), random vs planned arrivals (emergency vs appointments), heterogeneous demand sources (business vs leisure).
- Strategies: use off-peak capacity creatively (hotel conferences); complementary services (kids zone in cinema); **price incentives** (night data packs, mid-week discounts); **reservations** (benefits: customer — no wait, guaranteed service; firm — steadier demand, caps demand). Risk: reserved demand that does not show up = lost revenue → leads to overbooking (Part 3).

2.2 FORMULA BOXES (ALL FROM THE SHEET)

M/M/1/∞/∞ (λ < μ)

$$\rho = \lambda/\mu \quad P_0 = 1 - \rho \quad P_n = \rho^n P_0$$

$$L_S = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu-\lambda} \quad L_Q = \frac{\lambda^2}{\mu(\mu-\lambda)} \quad L_S = L_Q + \rho$$

$$W_S = \frac{1}{\mu-\lambda} \quad W_Q = \frac{\lambda}{\mu(\mu-\lambda)} \quad W_S = W_Q + 1/\mu$$

$$\text{Little: } L_S = \lambda W_S \quad L_Q = \lambda W_Q$$

EXTRAS DERIVED FROM P_N (NOT ON SHEET)

$$P(n \geq k) = \rho^k \quad P(n > k) = \rho^{k+1} \quad P(\text{server busy}) = \rho$$

$$\% \text{ time no queue} = P_0 + P_1 = (1-\rho)(1+\rho) \quad P(\text{wait} > 0) = \rho$$

M/M/1/N/∞ (MAX N IN SYSTEM; BLOCKED CUSTOMERS LOST)

$$\rho = \lambda/\mu \quad P_0 = \frac{1-\rho}{1-\rho^{N+1}} \quad P_n = \rho^n P_0 \quad P_N = \rho^N P_0 \text{ (blocking probability)}$$

$$\lambda_{\text{eff}} = \lambda(1 - P_N) \quad \text{lost} = \lambda P_N$$

$$L_S = \{ \rho \{ 1 + N \rho^{N+1} - (N+1) \rho^N \} / (1-\rho)(1-\rho^{N+1}) \}$$

$$L_Q = L_S - \lambda_{\text{eff}}/\mu \quad W_S = L_S/\lambda_{\text{eff}} \quad W_Q = L_Q/\lambda_{\text{eff}}$$

M/M/C/∞/∞ (C SERVERS, ONE QUEUE)

$$\rho = \frac{\lambda}{c\mu} \text{ (utilisation)} \quad r = \lambda/\mu = c\rho \text{ (offered load)}$$

$$P_0 = \frac{1}{\sum_{n=0}^{c-1} (c\rho)^n/n! + (c\rho)^c/[c!(1-\rho)]}$$

$$P_n = \frac{\lambda^n P_0}{\mu^n n!} \text{ for } n < c \quad P_n = \frac{\lambda^n P_0}{c! \mu^n c^{n-c}} \text{ for } n \geq c$$

$$L_Q = \frac{r^{c+1} P_0}{(c-r)^2 (c-1)!} \text{ [sheet writes } r \text{ as } \rho] \quad W_Q = L_Q/\lambda$$

$$W_S = W_Q + 1/\mu \quad L_S = \lambda W_Q + \lambda/\mu$$

$$P(\text{arrival must wait}) = \frac{r^c P_0}{c!(1-\rho)} \text{ (= } P(n \geq c), \text{ derived)}$$

M/D/1/∞/∞ (CONSTANT SERVICE TIME)

$$\rho = \lambda/\mu \quad P_0 = 1 - \rho$$

$$L_Q = \frac{\lambda^2}{2\mu(\mu-\lambda)} \quad W_Q = \frac{\lambda}{2\mu(\mu-\lambda)}$$

$$L_S = L_Q + \lambda/\mu \quad W_S = W_Q + 1/\mu$$

QUEUE COST (ECONOMICS OF WAITING)

Total cost/hr = (server cost/hr × no. of servers) + (waiting cost/hr per customer × L_Q or L_S)

Per customer: waiting cost = $c_w \times W_Q$ (or W_S if the problem charges time in system)

Read the wording: "waiting *in queue*" ⇒ use L_Q/W_Q . "Time in system" ⇒ L_S/W_S . Multiply hours by hours of operation for daily cost.

METHOD. 1) Identify model (Poisson/exp? fixed service? limit N? c servers?). 2) Write λ , μ , c per hour. 3) Check $\rho < 1$. 4) Compute ρ , then L_S/L_Q , then W via Little. 5) If costs given, build total cost for each option and compare. 6) State the decision in one sentence.

2.3 SOLVED EXAMPLES

EXAMPLE 2.1 — M/M/1 BASICS (CLASS: RAILWAY COUNTER)

EASY

Customers arrive every 3 minutes at a railway ticket counter; the teller serves 25 customers/hour. Infinite queue and population. (a) % of time teller idle, (b) % of time there is no queue. Also find L_Q and W_Q .

$$\lambda = 60/3 = 20/\text{hr}, \mu = 25/\text{hr} \Rightarrow \rho = 20/25 = 0.8.$$

$$(a) P_0 = 1 - \rho = 0.20 \rightarrow 20\%.$$

$$(b) \text{No queue} \Rightarrow 0 \text{ or } 1 \text{ in system: } P_0 + P_1 = 0.20 + 0.8(0.20) = 0.20 + 0.16 = 0.36 \rightarrow 36\%.$$

$$L_Q = \lambda^2 / [\mu(\mu - \lambda)] = 400 / (25 \times 5) = 3.2 \text{ customers. } W_Q = 20 / (25 \times 5) = 0.16 \text{ hr} = 9.6 \text{ min. } L_S = 3.2 + 0.8 = 4.0; W_S = 1/5 \text{ hr} = 12 \text{ min.}$$

(a) 20% (b) 36% $L_Q = 3.2$, $W_Q = 9.6 \text{ min}$

EXAMPLE 2.2 — M/M/1 PROBABILITIES (PHARMACY COUNTER)

EASY-MODERATE

Arrivals every 5 min; average service 4 min; one pharmacist. Find L_S , L_Q , W_S , W_Q , the probability that 3 or more people are in the system, and that more than 4 are.

$$\lambda = 12/\text{hr}, \mu = 60/4 = 15/\text{hr}, \rho = 0.8.$$

$$L_S = \rho / (1 - \rho) = 0.8 / 0.2 = 4. L_Q = 144 / (15 \times 3) = 3.2 \text{ (check: } 4 - 0.8 = 3.2 \text{ ✓).}$$

$$W_S = 1 / (15 - 12) = 1/3 \text{ hr} = 20 \text{ min. } W_Q = 12 / (15 \times 3) = 0.2667 \text{ hr} = 16 \text{ min (check: } L_Q / \lambda = 3.2 / 12 \text{ ✓).}$$

$$P(n \geq 3) = \rho^3 = 0.8^3 = 0.512. P(n > 4) = \rho^5 = 0.8^5 = 0.328.$$

$L_S=4$, $L_Q=3.2$, $W_S=20 \text{ min}$, $W_Q=16 \text{ min}$, $P(\geq 3)=0.512$, $P(>4)=0.328$

EXAMPLE 2.3 — FINITE QUEUE M/M/1/N (CLASS, THEN EXTENDED)

MODERATE

Same railway teller ($\lambda=20$, $\mu=25$) but at most $N = 10$ customers in the system. (a) % idle, (b) % no queue. Then interpret vs the infinite case.

$$\rho = 0.8. P_0 = (1 - \rho) / (1 - \rho^{11}) = 0.2 / (1 - 0.0859) = 0.2188 \rightarrow 21.9\%.$$

$$P_1 = \rho P_0 = 0.175; \text{no queue} = P_0 + P_1 = 0.394 \rightarrow 39.4\%.$$

Interpretation: idle time is *higher* with a finite queue (21.9% vs 20%; no-queue 39.4% vs 36%) because arrivals are turned away when full. Customer view: less congestion for admitted customers but risk of being blocked. Provider view: controls congestion and needs less waiting space, but lost customers and revenue.

21.9% and 39.4% (vs 20% and 36% for infinite queue)

EXAMPLE 2.4 — M/M/1/N WITH LOST REVENUE

DIFFICULT

A car-wash bay has room for at most $N = 4$ cars in total (one being washed). Cars arrive at $\lambda = 6/\text{hr}$; the wash rate is $\mu = 5/\text{hr}$. Each car brings Rs 200 contribution. Find P_0 , P_N , λ_{eff} , L_S , L_Q , W_S , W_Q , and lost contribution per 10-hour day.

$$\rho = 6/5 = 1.2 (>1 \text{ is allowed because the system is finite}).$$

$$P_0 = (1 - 1.2) / (1 - 1.2^5) = (-0.2) / (1 - 2.48832) = 0.2 / 1.48832 = 0.1344.$$

$$P_N = P_4 = 1.2^4 \times 0.1344 = 2.0736 \times 0.1344 = 0.2786 \text{ (27.9\% of arrivals are blocked).}$$

$$\lambda_{\text{eff}} = 6(1 - 0.2786) = 4.328/\text{hr. Lost} = 6 \times 0.2786 = 1.672/\text{hr.}$$

$$L_S = \rho \{1 + N\rho^{N+1} - (N+1)\rho^N\} / [(1-\rho)(1-\rho^{N+1})] = 1.2 \{1 + 4(2.48832) - 5(2.0736)\} / [(-0.2)(-1.48832)] = 1.2(1 + 9.9533 - 10.368) / 0.29766 = 1.2(0.5853) / 0.29766 = 2.359.$$

$$L_Q = L_S - \lambda_{\text{eff}}/\mu = 2.359 - 4.328/5 = 2.359 - 0.866 = \mathbf{1.494}.$$

$$W_S = 2.359/4.328 = 0.545 \text{ hr} = \mathbf{32.7 \text{ min.}} \quad W_Q = 1.494/4.328 = 0.345 \text{ hr} = \mathbf{20.7 \text{ min.}}$$

Lost contribution/day = $1.672 \times 10 \times 200 = \mathbf{Rs\ 3,344}$. Two decimals of ρ^{N+1} matter — keep 4 decimals.

$$\mathbf{P_N=0.279, \lambda_{\text{eff}}=4.33/\text{hr}, L_S=2.36, L_Q=1.49, W_S=32.7 \text{ min}, W_Q=20.7 \text{ min}, \text{loss} \approx \text{Rs } 3,344/\text{day}}$$

EXAMPLE 2.5 — FIRE ONE MECHANIC, HIRE ANOTHER? (CLASS: STAR GARAGE)

MODERATE

Raju installs 3 mufflers/hr; customers arrive at 1/hr; shop open 8 hr/day. Waiting cost = Rs 500 per hour of customer time in queue. Raju is paid Rs 150/hr. Shyam does 4/hr but is paid 50% more. Should Apte replace Raju by Shyam?

$\lambda = 1$. **Raju:** $\mu = 3 \Rightarrow L_Q = 1/[3(3-1)] = 1/6 = 0.1667$. Waiting cost/hr = $500 \times 0.1667 = 83.33$. Wage 150. Total/hr = 233.33 $\Rightarrow$ **Rs 1,866.67/day**.

Shyam: $\mu = 4 \Rightarrow L_Q = 1/[4(3)] = 1/12 = 0.0833$. Waiting cost/hr = 41.67. Wage = $1.5 \times 150 = 225$. Total/hr = 266.67 $\Rightarrow$ **Rs 2,133.33/day**.

(Same answer using W_Q : Raju $W_Q = 1/6$ hr; cost/hr = $500 \times \lambda \times W_Q = 83.33$ — identical because $L_Q = \lambda W_Q$ and $\lambda = 1$.)

Keep Raju: Rs 1,867 vs Rs 2,133 per day. Faster server does not pay for its wage premium.

EXAMPLE 2.6 — MULTI-SERVER M/M/C AND POOLING

DIFFICULT

A bank branch has $\lambda = 8$ customers/hr and each teller serves $\mu = 5/\text{hr}$. Compare (i) two tellers sharing **one** queue (M/M/2) with (ii) two tellers each with their own queue receiving 4/hr (two M/M/1). Then find M/M/3 as well.

(i) M/M/2: $\rho = 8/(2 \times 5) = 0.8$; $r = \lambda/\mu = 1.6$.

$$P_0 = 1/[(r^0/0!) + (r^1/1!) + r^2/(2!(1-\rho))] = 1/[1 + 1.6 + 2.56/(2 \times 0.2)] = 1/[1 + 1.6 + 6.4] = 1/9 = \mathbf{0.1111}.$$

$$L_Q = r^3 P_0 / [(2-1.6)^2 (1!)] = 4.096 \times 0.1111 / 0.16 = \mathbf{2.844}.$$

$$W_Q = L_Q/\lambda = 2.844/8 = 0.3556 \text{ hr} = \mathbf{21.3 \text{ min.}} \quad W_S = W_Q + 1/\mu = 0.3556 + 0.2 = 0.5556 \text{ hr} = \mathbf{33.3 \text{ min.}} \quad L_S = \lambda W_Q + \lambda/\mu = 2.844 + 1.6 = \mathbf{4.444}.$$

$$P(\text{customer has to wait}) = r^2 P_0 / [2!(1-\rho)] = 2.56 \times 0.1111 / 0.4 = \mathbf{0.711}.$$

(ii) Two M/M/1 with $\lambda = 4$: $W_Q = 4/[5(1)] = 0.8 \text{ hr} = \mathbf{48 \text{ min}}$; L_Q per line = 3.2. Pooling cuts the wait from 48 min to 21.3 min with the same 2 tellers.

(iii) M/M/3: $\rho = 8/15 = 0.5333$, $r = 1.6$. $P_0 = 1/[1 + 1.6 + 1.28 + 1.6^3/(6(1-0.5333))] = 1/[3.88 + 4.096/2.8] = 1/(3.88+1.4629) = \mathbf{0.1872}$. $L_Q = 1.6^4 \times 0.1872 / [(3-1.6)^2 \times 2!] = 6.5536 \times 0.1872 / 3.92 = \mathbf{0.3129}$. $W_Q = 0.3129/8 = 0.0391 \text{ hr} = \mathbf{2.35 \text{ min.}}$

M/M/2: $L_Q=2.84, W_Q=21.3 \text{ min}$ | 2×M/M/1: $W_Q=48 \text{ min}$ | M/M/3: $L_Q=0.31, W_Q=2.35 \text{ min}$

EXAMPLE 2.7 — CONSTANT SERVICE TIME M/D/1 (CLASS: RECYCLING FIRM)

MODERATE

Trucks currently wait 15 min (per trip). Waiting cost = Rs 6,000/hr. A new compactor serves at a constant 12 trucks/hr; arrivals are Poisson, 8/hr. Amortised cost of the compactor = Rs 300 per truck unloaded. Buy?

$$\text{Current waiting cost/truck} = 6000 \times (15/60) = \mathbf{Rs\ 1,500}.$$

$$\text{M/D/1: } \lambda = 8, \mu = 12. \quad W_Q = \lambda/[2\mu(\mu-\lambda)] = 8/[2 \times 12 \times 4] = 8/96 = 1/12 \text{ hr} = \mathbf{5 \text{ min.}}$$

$$\text{New waiting cost/truck} = 6000 \times (1/12) = \text{Rs } 500. \text{ Add amortised cost Rs } 300 \Rightarrow \mathbf{Rs\ 800/truck}.$$

$$\text{Saving} = 1500 - 800 = \text{Rs } 700 \text{ per truck.}$$

Check: the M/M/1 W_Q at the same rates = $8/[12 \times 4] = 10 \text{ min}$, exactly twice the M/D/1 value.

Buy the compactor: Rs 800 vs Rs 1,500 per truck (saves Rs 700).

EXAMPLE 2.8 — ECONOMICS OF WAITING: HOW MANY TELLERS? (CLASS: COOPERATIVE BANK)
MODERATE

Total waiting time accumulated per hour = $54/t^2$ minutes where t = number of tellers. Waiting costs Rs 10/min; a teller costs Rs 40/hr. (1) Optimal tellers? (2) Show the trade-off from $t-2$ to $t+2$. (3) Over- or under-capacitate?

$$\text{Total cost}(t) = 40t + 10 \times 54/t^2 = 40t + 540/t^2.$$

t	Teller cost	Waiting cost	Total
1	40	540	580
2	80	135	215
3	120	60	180
4	160	33.75	193.75
5	200	21.60	221.60

Check with calculus: $d/dt = 40 - 1080/t^3 = 0 \Rightarrow t^3 = 27 \Rightarrow t = 3$.

Trade-off: as t rises capacity cost climbs linearly while waiting cost falls steeply then flattens $\Rightarrow$ U-shaped total cost $\checkmark$. Being **2 below** the optimum costs +35 (215); **2 above** costs +41.6 (221.6) — but **1 above** costs only +13.75 vs **1 below** +35.

3 tellers (Rs 180/hr). Prefer slightly over-capacity: the cost curve is steeper on the under-capacity side.

EXAMPLE 2.9 — TWO-PHASE QUEUE: HIRE A SECOND REGISTRATION CLERK? (CLASS SCENARIO)
DIFFICULT

Patients arrive at 10/hr. Phase 1 registration: one clerk, $\mu_1 = 12$ /hr. Phase 2 doctor: $\mu_2 = 15$ /hr. Patient waiting cost = Rs 150/hr; clerk Rs 300/hr; doctor Rs 1,200/hr. Clinic works 8 hr/day. Is a second registration clerk worthwhile (two clerks working as M/M/2 each at 12/hr)?

Doctor phase is unchanged in both options, so compare only Phase 1. (Departures from an M/M/1 queue are again Poisson at $\lambda = 10$, so the phases can be analysed separately.)

Option A — one clerk (M/M/1): $L_S = 10/(12-10) = 5$ patients; $L_Q = 100/(12 \times 2) = 4.167$; $W_S = 30$ min. Cost/hr = $150 \times 5 + 300 = \text{Rs } 1,050$.

Option B — two clerks (M/M/2): $\rho = 10/24 = 0.4167$; $r = 0.8333$. $P_0 = 1/[1 + 0.8333 + 0.8333^2/(2 \times 0.5833)] = 1/[1.8333 + 0.5952] = 0.4118$. $L_Q = 0.8333^3 \times 0.4118 / [(2 - 0.8333)^2 \times 1] = 0.5787 \times 0.4118 / 1.3611 = 0.1751$. $W_Q = 1.05$ min. $L_S = 0.1751 + 0.8333 = 1.0084$; $W_S = 6.05$ min. Cost/hr = $150 \times 1.0084 + 600 = \text{Rs } 751.3$.

Saving = $1,050 - 751.3 = \text{Rs } 298.7/\text{hr} \approx \text{Rs } 2,390/\text{day}$. (Using L_Q instead of L_S gives the same saving, because $L_S - L_Q = \lambda/\mu$ is identical for both.)

Yes, hire the second clerk: saves $\approx$ Rs 299/hr ($\approx$ Rs 2,390/day); patient time in registration drops from 30 min to 6 min.

Simulation-style question: compute waiting times from an arrival table (class practice)
EXAMPLE 2.10 — WAITING FROM A TABLE (CLASS PRACTICE A-I)
MODERATE

Arrivals (clock minutes after 08:00) and service times: A 08:01 (2), B 08:07 (10), C 08:09 (4), D 08:20 (13), E 08:31 (9), F 08:39 (2), G 08:40 (7), H 08:52 (5), I 08:54 (3). Find each customer's wait and the total for (a) one teller, (b) two tellers.

Rule: **start = max(arrival, time server becomes free)**; wait = start – arrival; free-time = start + service. With two tellers send the customer to whichever is free first.

Cust	Arrive	Serv	Start (1 teller)	Wait	Leaves	Start (2 tellers)	Wait
A	08:01	2	08:01	0	08:03	08:01	0
B	08:07	10	08:07	0	08:17	08:07	0
C	08:09	4	08:17	8	08:21	08:09	0
D	08:20	13	08:21	1	08:34	08:20	0
E	08:31	9	08:34	3	08:43	08:31	0
F	08:39	2	08:43	4	08:45	08:39	0
G	08:40	7	08:45	5	08:52	08:40	0
H	08:52	5	08:52	0	08:57	08:52	0
I	08:54	3	08:57	3	09:00	08:54	0

Total workload = 2+10+4+13+9+2+7+5+3 = **55 min**; capacity 60 min (1 teller) or 120 min (2 tellers).

One teller: total wait = 24 min (8+1+3+4+5+0+3). Two tellers: total wait = 0.

Try yourself: redo Example 2.6 with $\lambda = 12/\text{hr}$, $\mu = 5/\text{hr}$ for $c = 3$ and $c = 4$ and find the cheapest c if a teller costs Rs 200/hr and waiting costs Rs 300 per customer-hour in the queue.

Revenue & Yield Management

Selling the right product to the right customer at the right time, through the right channel, at the right price.

3.1 CONCEPTS

- **Definition:** using pricing to increase the profit generated from limited (perishable) assets.
- **Works best when:** value differs across segments (business vs vacationers); the product is perishable (seats, rooms); demand has seasonality/peaks; services are sold in advance. Core dilemma: *accept an early discounted booking or hold the unit for a late high-paying customer?*
- **Traditional pricing** (Total revenue = $P \times Q$ on one demand curve) ignores segments with inelastic demand (business traveller booking last minute).
- **Fare classes (airlines):** Corporate (buy any time, fully refundable/changeable), Vacationers (≥ 2 weeks advance, non-refundable), Internet special (when flight not expected full, non-refundable).
- **Two questions RM answers:** how to differentiate segments so one pays more, and how to stop the low-fare segment consuming all capacity ($\rightarrow$ **capacity protection**).
- **Overbooking** applies when capacity is fixed & perishable and customers cancel/no-show. Denied service $\rightarrow$ airlines compensate/re-accommodate; hotels re-accommodate at other hotels; air cargo re-books; clinics use overtime.

3.2 FORMULA BOXES

CRITICAL RATIO (NEWSVENDOR LOGIC)

C_u = cost of *under*-protecting/under-booking C_o = cost of *over*-protecting/over-booking

$$CR = \frac{C_u}{C_u + C_o} \quad Q^* = \mu + z\sigma \text{ where } P(X \leq Q^*) = CR \text{ (z from normal table)}$$

LITTLEWOOD'S RULE (CAPACITY PROTECTION FOR THE HIGH FARE)

Protect the next seat for the high fare while $F_H \times P(D_H > Q) > F_L$

$$\text{At the optimum: } P(D_H > Q^*) = \frac{F_L}{F_H} \Leftrightarrow P(D_H \leq Q^*) = \frac{F_H - F_L}{F_H} = CR$$

$C_u = F_H - F_L$ (protected too few $\rightarrow$ sold a seat cheap that could earn F_H)

$C_o = F_L$ (protected too many $\rightarrow$ seat empty, could have earned F_L)

$Q^* = \mu_H + z\sigma_H$ = seats to **protect** for high fare $\Rightarrow$ max low-fare seats to sell = Capacity - Q^*

$\sigma = \sqrt{\text{variance}}$. "CR probability = 1 - Littlewood probability": Littlewood probability = $P(D_H > Q) = F_L/F_H$.

OVERBOOKING

Let X = number of no-shows (or no-show %). Overbook by q where $P(X \leq q) \geq CR$

C_u = cost when a booked customer **doesn't show** and the room/seat goes empty (lost contribution)

C_o = cost when someone **shows and is denied** (compensation + goodwill)

$$CR = \frac{C_u}{C_u + C_o} \quad q^* = \mu_{\text{no-show}} + z\sigma_{\text{no-show}} \text{ (normal) or smallest } q \text{ with cumulative } P(X \leq q) \geq CR \text{ (discrete table)}$$

MULTI-SEGMENT PRICING — UNCAPACITATED

Demand of segment i : $d_i = A_i - B_i p_i$ unit cost c

$$\text{Profit} = \sum (p_i - c)(A_i - B_i p_i)$$

$$\text{Optimal price: } p_i^* = \frac{A_i}{2B_i} + \frac{c}{2} \quad (\text{set } d(\text{profit})/dp = 0)$$

MULTI-SEGMENT PRICING — CAPACITATED (TOTAL CAPACITY Q)

$$\text{Max } \sum (p_i - c)(A_i - B_i p_i) \quad \text{s.t.} \quad \sum (A_i - B_i p_i) \leq Q, \quad A_i - B_i p_i \geq 0$$

Step 1: compute uncapacitated demand. If $\sum d_i \leq Q \rightarrow$ constraint slack, use the uncapacitated prices.

Step 2 (binding): replace c by $s = c + \lambda$ (shadow price): $p_i = \frac{A_i}{2B_i} + \frac{s}{2}$. Demand $d_i = \frac{A_i}{2} - \frac{B_i s}{2}$

$$\text{Solve } \sum d_i = Q: \quad s = \frac{\sum A_i/2 - Q}{\sum B_i/2} \Rightarrow \text{shadow price } \lambda = s - c$$

Profit computed with the *true* cost c (not s).

Same-price (single segment) benchmark: add demand curves $d = \sum A - (\sum B)p$ and apply $p = A/(2B) + c/2$ to the aggregate.

3.3 SOLVED EXAMPLES

EXAMPLE 3.1 — LITTLEWOOD (CLASS: BUDGET AIRLINE)

MODERATE

Full-fare and internet-special return tickets cost Rs 6,900 and Rs 4,900. Aircraft has 95 seats; ample demand for the internet special. Full-fare demand $\sim$ Normal(mean 60, variance 225). How many internet-special tickets should be sold at most?

$$F_H = 6900, F_L = 4900. C_U = 2000, C_O = 4900. CR = 2000/6900 = \mathbf{0.2899}.$$

$$\sigma = \sqrt{225} = 15. \text{ Find } z \text{ with } P(Z \leq z) = 0.2899 \Rightarrow z \approx \mathbf{-0.55} \text{ (z-table: 0.2912 at } -0.55; \text{ exact } -0.554).$$

$$Q^* = 60 + (-0.554)(15) = 60 - 8.3 = \mathbf{51.7} \rightarrow \text{protect } \mathbf{52} \text{ seats for full fare.}$$

$$\text{Max internet specials} = 95 - 52 = \mathbf{43}.$$

Sanity check: $F_L/F_H = 0.71 \Rightarrow$ we protect only until there is a 71% chance of selling the marginal seat at full fare, so Q^* is below the mean ($51.7 < 60$).

Protect ≈ 52 seats for full fare; sell at most 43 internet-special tickets.

EXAMPLE 3.2 — HOTEL PROTECTION LEVEL (CLASS)

MODERATE

Hotel has 120 rooms. Corporate Flex Rate Rs 10,000; Advance Saver Rs 2,000 (sells out). Flex demand $\sim$ Normal(70, variance 225). How many rooms to protect for Flex and what is the max number of Advance Saver bookings?

$$CR = (10000 - 2000)/10000 = \mathbf{0.80} \Rightarrow z = \mathbf{+0.84}. \sigma = 15.$$

$$Q^* = 70 + 0.84(15) = 70 + 12.6 = \mathbf{82.6} \rightarrow \text{protect } \mathbf{83} \text{ rooms.}$$

$$\text{Advance Saver limit} = 120 - 83 = \mathbf{37} \text{ rooms.}$$

Note the direction: high fare is $\sim 5\times$ the low fare, so the hotel protects *more* than the mean ($z > 0$). Always check the sign of z against F_L/F_H : if $F_L/F_H < 0.5$ then $Q^* > \mu$.

Protect 83 rooms for Flex; accept at most 37 Advance Saver bookings.

EXAMPLE 3.3 — OVERBOOKING WITH NORMAL NO-SHOWS (CLASS HOTEL)

MODERATE

Historically 30% no-shows, variance 9 ($\%^2$, so $\sigma = 3\%$). Cost of an empty room from a no-show = Rs 3,200; cost when a booked guest shows and there is no room = Rs 8,000. Extent of overbooking?

$C_u = 3200$ (no-show $\rightarrow$ empty room), $C_o = 8000$ (denied guest). $CR = 3200/11200 = 0.2857 \Rightarrow z = -0.57$.

$q^* = 30\% + (-0.57)(3\%) = 30\% - 1.7\% = \approx 28.3\%$.

Interpretation: plan overbooking around a 28.3% no-show level (class treatment: overbooking extent $q^* = 28.3\%$). Being cautious (below the 30% mean) is right because a denied guest costs 2.5 $\times$ an empty room. If the exam gives a room count, state your reading: e.g. covering 28.3% expected no-shows on N booked rooms means accepting N bookings for $N(1-0.283)$ rooms of capacity.

Be ready for the exam wording: if variance were given as 0.09 for a proportion, $\sigma = 0.3$ — state your interpretation.

Overbook to about 28.3% no-shows ($z \approx -0.57$): a conservative level because $C_o > C_u$.

EXAMPLE 3.4 — OVERBOOKING FROM A DISCRETE TABLE (CLASS)

MODERATE

No-show distribution $P(d)$: 0 $\rightarrow$.07, 1 $\rightarrow$.19, 2 $\rightarrow$.22, 3 $\rightarrow$.16, 4 $\rightarrow$.12, 5 $\rightarrow$.10, 6 $\rightarrow$.07, 7 $\rightarrow$.04, 8 $\rightarrow$.02, 9 $\rightarrow$.01. $C_u = \$40$ (room contribution lost when a reservation is not honoured, i.e., no-shows underestimated). $C_o = \$100$ (cost of not having a room for an overbooked guest). How many reservations to overbook?

$CR = 40/(40+100) = 0.2857$.

q	0	1	2	3	4	5	6
P(d)	.07	.19	.22	.16	.12	.10	.07
Cumulative P(d $\leq$ q)	.07	.26	.48	.64	.76	.86	.93

Smallest q with cumulative ≥ 0.2857 is q = 2 ($0.26 < 0.2857 \leq 0.48$).

Cross-check by marginal analysis: the 2nd overbooked booking is worth it if $40 \times P(\text{no-shows} \geq 2) > 100 \times P(\text{no-shows} < 2)$: $40(0.74) = 29.6 > 100(0.26) = 26 \checkmark$. The 3rd: $40(0.52) = 20.8 < 100(0.48) = 48 \times$.

Overbook by 2 reservations.

EXAMPLE 3.5 — TWO-SEGMENT PRICING (CLASS)

MODERATE

$d_1 = 5000 - 20p_1$; $d_2 = 5000 - 40p_2$; $c = \text{Rs } 10$. (a) Optimal prices and total profit. (b) What if one single price is forced? (c) Capacity $Q = 4,000$.

(a) $p_1 = 5000/(2 \times 20) + 10/2 = 125 + 5 = \text{Rs } 130$; $p_2 = 5000/80 + 5 = 62.5 + 5 = \text{Rs } 67.5$.

$d_1 = 5000 - 2600 = 2,400$; $d_2 = 5000 - 2700 = 2,300$ (total 4,700). Profit = $(130-10)(2400) + (67.5-10)(2300) = 288,000 + 132,250 = \text{Rs } 420,250$.

(b) Aggregate demand $d = 10000 - 60p \Rightarrow p = 10000/120 + 5 = \text{Rs } 88.33$; $d = 4,700$; profit = $(78.33)(4700) = \text{Rs } 368,167$. Price discrimination gains $420,250 - 368,167 = \text{Rs } 52,083$ (+14.1%).

(c) Uncapacitated demand $4,700 > 4,000 \Rightarrow$ constraint binds. $s = (\sum A/2 - Q)/(\sum B/2) = (5000 - 4000)/30 = 33.33$ (so $\lambda = s - c = 23.33$ per unit).

$p_1 = 125 + 16.67 = \text{Rs } 141.67$; $p_2 = 62.5 + 16.67 = \text{Rs } 79.17$. $d_1 = 5000 - 20(141.67) = 2,166.7$; $d_2 = 5000 - 40(79.17) = 1,833.3$; total = 4,000 $\checkmark$.

Profit = $(141.67-10)(2166.7) + (79.17-10)(1833.3) = 285,278 + 126,806 = \text{Rs } 412,083$. Both prices rise; the capacity limit costs Rs 8,167 relative to unlimited capacity.

(a) Rs 130 / Rs 67.5, profit Rs 420,250 (b) single price Rs 88.33, profit Rs 368,167 (c) Rs 141.67 / Rs 79.17, profit Rs 412,083

EXAMPLE 3.6 — THREE-SEGMENT CAPACITATED PRICING

DIFFICULT

A conference venue sells to three segments: $d_1 = 3000 - 10p_1$, $d_2 = 2000 - 10p_2$, $d_3 = 1500 - 5p_3$. Cost per unit $c = 20$. Capacity = 2,500 units. Find prices, allocation, profit, and the value of one extra unit of capacity.

Uncapacitated: $p_1 = 3000/20 + 10 = 160$; $p_2 = 2000/20 + 10 = 110$; $p_3 = 1500/10 + 10 = 160$. Demands $1,400 + 900 + 700 = 3,000 > 2,500 \rightarrow$ **binds**.

$\sum A/2 = (3000+2000+1500)/2 = 3250$; $\sum B/2 = (10+10+5)/2 = 12.5$. $s = (3250 - 2500)/12.5 = 60 \Rightarrow \lambda = 60 - 20 = 40$.

Prices: $p_1 = 150 + 30 = 180$; $p_2 = 100 + 30 = 130$; $p_3 = 150 + 30 = 180$.

Demands: $d_1 = 3000 - 1800 = 1,200$; $d_2 = 2000 - 1300 = 700$; $d_3 = 1500 - 900 = 600$. Total = 2,500 ✓.

Profit = $(180-20)(1200) + (130-20)(700) + (180-20)(600) = 192,000 + 77,000 + 96,000 = 365,000$ (vs 375,000 if capacity were unlimited).

The shadow price $\lambda = 40$ says one extra unit of capacity is worth about 40 of extra profit.

p = 180 / 130 / 180; allocation 1,200 / 700 / 600; profit 365,000; capacity worth ≈ 40 per unit.

EXAMPLE 3.7 — PROTECTION LEVEL WITH A VERY LOW DISCOUNT FARE (VARIATION)

MODERATE-DIFFICULT

Flight of 150 seats. $F_H = \text{Rs } 12,000$, $F_L = \text{Rs } 3,000$. High-fare demand $\sim \text{Normal}(\mu = 45, \sigma = 12)$. (a) Seats to protect; (b) max discount seats; (c) what happens to Q^* if F_L rises to Rs 8,000?

(a) $CR = (12000-3000)/12000 = 0.75 \Rightarrow z = 0.67$. $Q^* = 45 + 0.67(12) = 45 + 8.04 = 53.0 \rightarrow$ **protect 53**.

(b) Discount seats = $150 - 53 = 97$.

(c) $CR = (12000-8000)/12000 = 0.333 \Rightarrow z = -0.43$. $Q^* = 45 - 0.43(12) = 45 - 5.2 = 39.8 \rightarrow 40$. When the discount fare gets closer to the full fare, protecting a seat is less attractive, so we protect fewer seats.

(a) 53 (b) 97 (c) protect 40 (fewer)

3.4 COMPETITION IN REVENUE MANAGEMENT (GAME THEORY: CONCEPTS)

Approach	When to use	Pros	Cons	Examples
Nash (simultaneous)	Symmetric/fragmented market; move at the same time; transparent prices (OTAs); no clear leader	Stable equilibrium; fair share; easy	Thin margins; reactive; no leadership edge	Uber vs Ola surge, Swiggy vs Zomato, budget hotels on OTAs
Stackelberg (leader–follower)	Clear leader; first-mover advantage; followers must adapt (residual demand)	Leader captures premium demand, anticipates follower; higher leader profit	Followers earn less; needs foresight; leader may miscalculate	Marriott/Taj advance rates, IndiGo hubs, McKinsey fees, Jio entry
Collaboration	High fixed & perishable capacity; few large players; price wars destroy value	Higher joint profits; stable pricing	Needs trust & enforcement; may reduce innovation	Airline alliances, tower sharing, rate-parity agreements, referral networks

- **Nash:** both set prices/room allocations at the same time without seeing the other's choice; equilibrium = neither can improve by changing alone.
- **Stackelberg:** the leader sets first (e.g., advance-booking protection levels), the follower adjusts optimally; the leader shapes residual demand.

Service Facility Layout (slide 47 onwards)

Product layout, process layout, line balancing, bottlenecks, labour utilisation and cost of capacity.

4.1 CONCEPTS

- **Product layout** (line flow, like a vehicle assembly line: chassis drop → engine mounting → cab mounting → ...). Used for **high volume, low variety**: standardised service to many customers (cafeteria line, airport security). Customer is an input moving through stations.
- **Process layout**: functional departments; customers route differently (retail shop, hospital) → for high variety, low volume.
- **McDonald's innovations**: indoor seating (1950s), drive-through (1970s), breakfast (1980s), play areas (late 1980s), self-service kiosk (2004). **4 of 5 concern layout**.
- **Service design factors**: (a) objective of service (emergency ward: no traffic, easy approach; petrol pump: bright colour visible from afar); (b) space requirement & location (rural vs urban → vertical expansion); (c) security (CCTV, biometrics at immigration); (d) flexibility (adapt to new equipment, expansion provision).
- **Mongolian Grill**: customers assemble their own bowls at open-grill/food-prep stations. Objective: raise capacity and throughput with self-service. Result: higher profit; small bottlenecks (≤ 2 min) cleared by cooks directing traffic; repeat customers who understand queues walk to the *farthest* prep area first and reduce bottlenecks — customer behaviour affects service quality/flow.
- **Bottleneck strategy**: add parallel workers/equipment at the slowest station, or re-split the tasks, to balance the line — but every extra worker adds cost, so compare **cost per customer**.

4.2 FORMULA BOXES (FROM THE SHEET, PAGE 3)

LINE BALANCING FORMULAS

$$\text{Capacity (per station)} = \frac{\text{Available time}}{\text{Processing time}} \quad \text{e.g. } 3600 \text{ s/hr} \div \text{time per customer}$$

$$\text{Station capacity} = \text{No. of parallel workers} \times \text{individual capacity} \quad \text{Effective processing time} = \frac{\text{Processing time}}{\text{No. of parallel workers}}$$

$$\text{Process capacity} = \text{capacity of the bottleneck} = \min \text{ over stations} = 3600 / \max(\text{effective time}) (= 3600/\text{cycle time})$$

$$\text{Ideal cycle time} = \frac{\text{Labour content}}{\text{No. of workers}} \quad (\text{perfectly balanced}) \quad \text{Labour content} = \text{sum of all task times}$$

$$\text{Labour utilisation (\%)} = \frac{\text{Labour content}}{\text{Cycle time} \times \text{No. of workers}} \times 100 \quad (\text{cycle time} = \text{bottleneck time})$$

$$\text{Cost of capacity: cost per customer} = \frac{(\text{No. of workers}) \times (\text{wage per hr})}{\text{Process capacity per hr}}$$

METHOD. 1) Convert times to a common unit (seconds). 2) Effective time of each station = time ÷ workers. 3) Bottleneck = largest effective time ⇒ cycle time. 4) Capacity/hr = 3600/cycle time. 5) Utilisation = labour content / (cycle time × total workers). 6) Cost per customer = workers × wage / capacity. 7) Recommend on **cost per customer** and whether capacity meets demand.

4.3 SOLVED EXAMPLES

EXAMPLE 4.1 — CAFETERIA LINE (CLASS)

MODERATE

Four stations, 1 worker each: drinks 20 s, starters & soup 30 s, main course 40 s, dessert 15 s. Lunch window 1 hr. (a) Capacity and utilisation as-is. (b) Ideal balanced line? (c) Option: double the main-course station.

Labour content = 20+30+40+15 = **105 s**. 4 workers.

(a) **As-is:** bottleneck = main course 40 s $\Rightarrow$ cycle time 40 s $\Rightarrow$ capacity = $3600/40 = 90$ customers/hr. Station capacities: 180, 120, 90, 240. Utilisation = $105/(40 \times 4) = 65.63\%$.

(b) **Ideal:** cycle = $105/4 = 26.25$ s each $\Rightarrow$ capacity $3600/26.25 = 137.14$ /hr, utilisation **100%** (theoretical; tasks rarely divide perfectly).

(c) **Two workers at main course** (effective 20 s): times 20, 30, 20, 15 $\Rightarrow$ bottleneck 30 s (starters) $\Rightarrow$ capacity = **120/hr**; 5 workers: utilisation = $105/(30 \times 5) = 70\%$.

Cost effect: workers 4 $\rightarrow$ 5 (+25%), capacity 90 $\rightarrow$ 120 (+33%) $\Rightarrow$ cost per customer falls ($4X/90 = 0.0444X$ vs $5X/120 = 0.0417X$). Next bottleneck is starters — add a worker there only if demand > 120 /hr.

As-is: 90/hr, 65.63%. Ideal: 137.14/hr, 100%. With 2 at main: 120/hr, 70%, cheaper per customer.

EXAMPLE 4.2 — AIRPORT SECURITY CHECKPOINT (CLASS: PROPOSAL QUESTION)

DIFFICULT

Four sequential activities, 1 employee each at Rs 400/hr: document verification 24 s, baggage screening 38 s, security check 40 s, exit verification 20 s. Evaluate these proposals: (P1) add one more employee at security check; (P2) add one more at baggage screening **and** one more at security check. Should management implement?

Base: labour content = $24+38+40+20 = 122$ s; bottleneck 40 s $\Rightarrow$ capacity $3600/40 = 90$ /hr; 4 workers cost 1,600/hr $\Rightarrow$ **Rs 17.78 per passenger**; utilisation = $122/(40 \times 4) = 76.25\%$.

P1 (security check $\times 2$): effective times 24, 38, 20, 20 $\Rightarrow$ new bottleneck = baggage 38 s $\Rightarrow$ capacity = $3600/38 = 94.74$ /hr; 5 workers = 2,000/hr $\Rightarrow$ **Rs 21.11/passenger** (worse); utilisation = $122/(38 \times 5) = 64.2\%$. **Reject P1** — only a 5% capacity gain for 25% more labour.

P2 (baggage $\times 2$ and security $\times 2$): effective times 24, 19, 20, 20 $\Rightarrow$ bottleneck = document verification 24 s $\Rightarrow$ capacity = $3600/24 = 150$ /hr; 6 workers = 2,400/hr $\Rightarrow$ **Rs 16.00/passenger**; utilisation = $122/(24 \times 6) = 84.72\%$. **Accept P2:** +67% capacity for +50% labour, and cheaper per passenger.

Extra: adding a 7th worker at document verification (12 s effective) leaves bottleneck at 20 s $\Rightarrow$ 180/hr, 7 workers $\Rightarrow$ Rs 15.56/pax, utilisation 87.14% — still cheaper per passenger if demand exists.

Reject P1 (Rs 21.11/pax); implement P2 (Rs 16.00/pax, 150/hr vs 90/hr). Decide with cost per customer AND demand.

EXAMPLE 4.3 — SIZING WORKERS TO MEET DEMAND

MODERATE-DIFFICULT

A three-station service line: verification 30 s, processing 45 s, delivery 20 s. Target throughput = 100 customers/hr. How many workers per station? Capacity and utilisation?

Required cycle time = $3600/100 = 36$ s per customer.

Workers needed per station = $\lceil \text{time} / 36 \rceil$: verification $\lceil 30/36 \rceil = 1$; processing $\lceil 45/36 \rceil = 2$ (effective 22.5 s); delivery $\lceil 20/36 \rceil = 1$. Total workers = 4.

Effective times 30, 22.5, 20 $\Rightarrow$ bottleneck = verification 30 s $\Rightarrow$ capacity = $3600/30 = 120$ /hr (≥ 100 $\checkmark$).

Labour content = $30+45+20 = 95$ s. Utilisation = $95/(30 \times 4) = 79.17\%$. At the target of 100/hr (36 s cycle), labour actually needed = $95/36 = 2.64$ workers, so idle labour is unavoidable with discrete workers.

1 + 2 + 1 workers; capacity 120/hr; utilisation 79.2%.

EXAMPLE 4.4 — WHICH LAYOUT, AND WHY?

CONCEPT

Why did the cafeteria use a product layout, and how does it differ from a retail shop?

Cafeteria: standard service, high volume in a 1-hour lunch window $\rightarrow$ stations arranged in the sequence of service (drinks $\rightarrow$ starters $\rightarrow$ main $\rightarrow$ dessert); efficiency comes from balancing station times; the *customer is an input* moving along the line. Retail shop: many product categories and customers choose their own route $\rightarrow$ process (functional) layout; high variety, low volume.

High volume + low variety $\Rightarrow$ product layout; high variety + low volume $\Rightarrow$ process layout.

Exam Plan & Checklists

2-HOUR TIME PLAN (ADAPT TO MARKS)

Minutes	What to do
0–5	Read the whole paper. Mark easy points. Note which model each queue question is.
5–55	Queuing questions (biggest weight). Write givens in per-hour units first.
55–90	Revenue questions. Identify CR, z , σ (root of variance), then the number. For pricing: $p = A/2B + c/2$, check capacity binding.
90–110	Layout questions: effective times → bottleneck → capacity → utilisation → cost per customer.
110–120	Theory answers + re-check units, rounding, and that each answer states a decision .

STANDARD PITFALLS

- ☐ λ and μ in the same unit and **per hour**. “Every 3 min” is arrival *interval* $\Rightarrow \lambda = 20/\text{hr}$.
- ☐ Check $\rho < 1$ for infinite-queue models. If $\rho \geq 1$, say the queue grows without bound.
- ☐ M/M/1/N: N counts the customer in service. Use λ_{eff} for W . ρ may exceed 1.
- ☐ M/M/c: P_0 uses $\sum$ up to $c-1$ and the last term with $(1-\rho)$; L_Q uses $r = \lambda/\mu$.
- ☐ “% time no queue” = $P_0 + P_1$, not P_0 alone.
- ☐ Cost questions: waiting cost uses L_Q if time *in queue*, L_S if *in system*. Compare on the same basis (per hour or per day).
- ☐ Littlewood: Q^* is seats **protected** for high fare; discount seats = capacity – Q^* . Use $\sigma = \sqrt{\text{variance}}$.
- ☐ Overbooking: identify which cost is C_u (empty room / no-show) and which is C_o (denied guest). Swapping them flips the answer.
- ☐ Pricing: check whether capacity binds by summing the uncapacitated demands first. Profit uses true c .
- ☐ Layout: cycle time = bottleneck *effective* time; utilisation denominator uses total workers (including extra ones).
- ☐ Round protection levels/workers sensibly and state the rounding; always give a one-line recommendation.

ONE-PAGE FORMULA RECAP

Topic	Key formulas
M/M/1	$\rho = \lambda/\mu$; $P_0 = 1 - \rho$; $P_n = \rho^n P_0$; $L_S = \lambda/(\mu - \lambda)$; $L_Q = \lambda^2/[\mu(\mu - \lambda)]$; $W_S = 1/(\mu - \lambda)$; $W_Q = \lambda/[\mu(\mu - \lambda)]$; $L = \lambda W$
M/M/1/N	$P_0 = (1 - \rho)/(1 - \rho^{N+1})$; $P_N = \rho^N P_0$; $\lambda_{\text{eff}} = \lambda(1 - P_N)$; $L_Q = L_S - \lambda_{\text{eff}}/\mu$; $W = L/\lambda_{\text{eff}}$
M/M/c	$\rho = \lambda/(c\mu)$; $r = \lambda/\mu$; P_0 formula; $L_Q = r^{c+1} P_0 / [(c - r)^2 (c - 1)!]$; $W_Q = L_Q/\lambda$; $W_S = W_Q + 1/\mu$; $L_S = \lambda W_Q + \lambda/\mu$
M/D/1	$L_Q = \lambda^2/[2\mu(\mu - \lambda)]$; $W_Q = \lambda/[2\mu(\mu - \lambda)]$; $L_S = L_Q + \lambda/\mu$; $W_S = W_Q + 1/\mu$
Queue economics	Total cost = server cost + waiting cost ($c_w \times L$ or W); choose minimum
Littlewood / CR	$C_u = F_H - F_L$; $C_o = F_L$; $CR = (F_H - F_L)/F_H$; $Q^* = \mu + z\sigma$; discount seats = $\text{Cap} - Q^*$
Overbooking	$CR = C_u/(C_u + C_o)$, C_u = empty-room loss, C_o = denied-guest cost; $q^* = \mu + z\sigma$ or smallest q with cum. $P \geq CR$
Pricing	$p = A/(2B) + c/2$; capacitated: $s = (\sum A/2 - Q)/(\sum B/2)$, $p = A/(2B) + s/2$, $\lambda = s - c$
Layout	Capacity = time/processing time; eff. time = $t/\text{workers}$; cycle = max eff. time; ideal cycle = labour content/workers; util = $LC/(\text{cycle} \times \text{workers})$